INCH-POUND

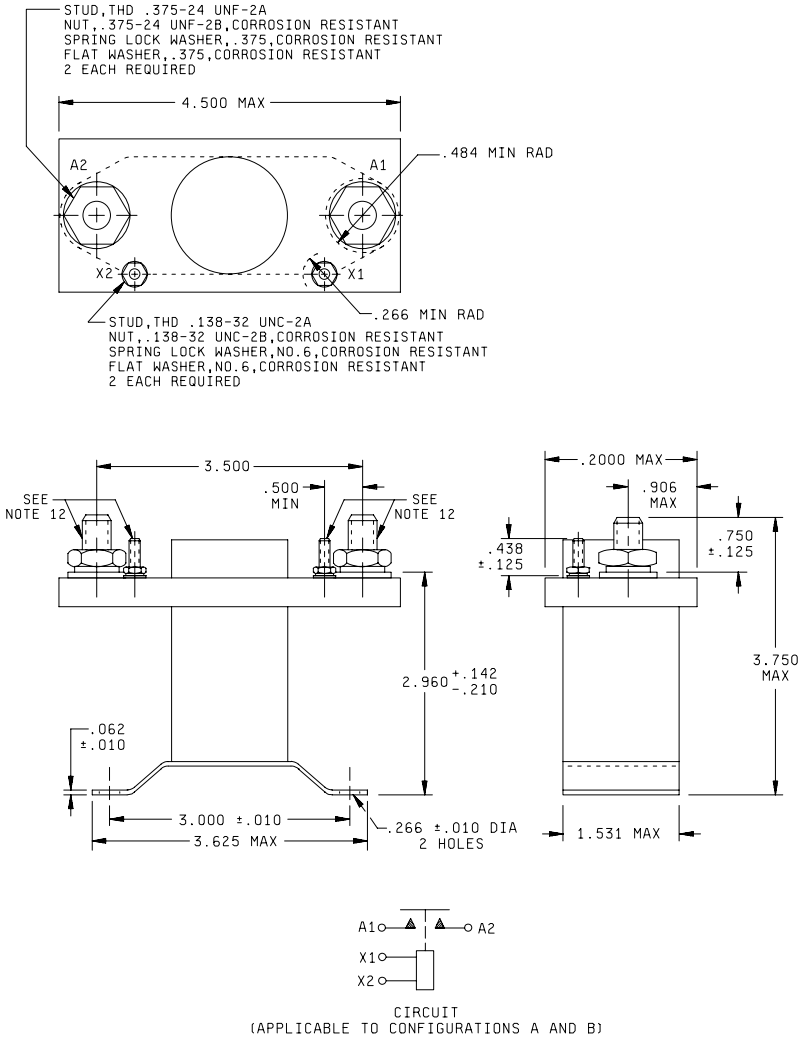
MS24172J
w/AMENDMENT 1
15 October 2004
SUPERSEDING
MS24172J
15 April 2003

DETAIL SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 200 AMPERES, 1 PST (N.O.), TYPE III, NONHERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

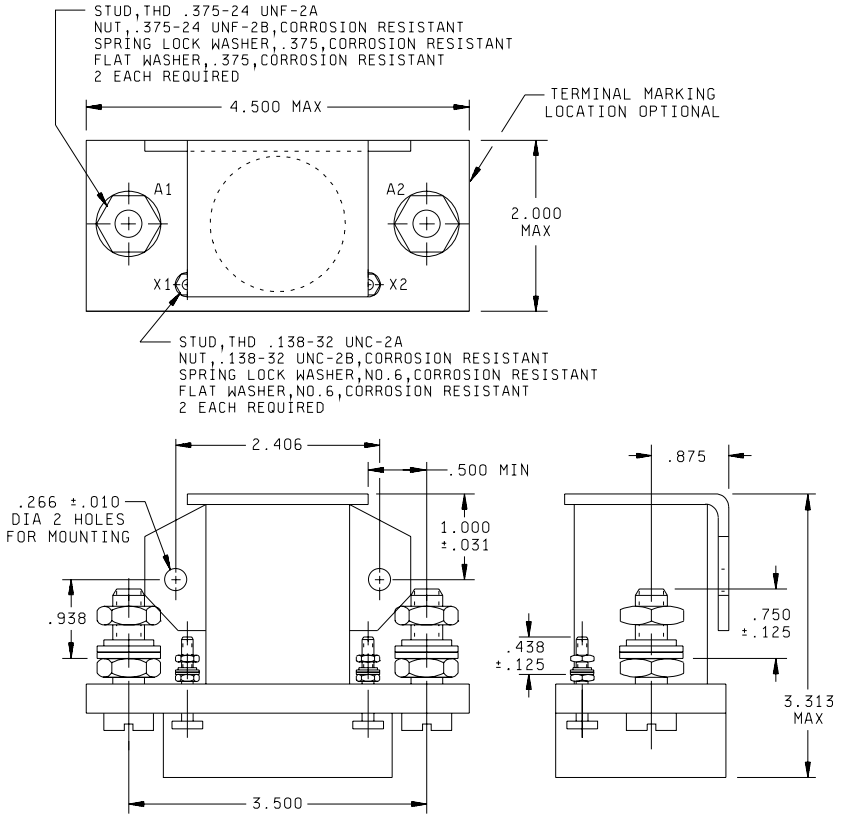
The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.



CONFIGURATION A

FIGURE I. Outline drawing (for details see tables I and II).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:49Z
Check the source to verify that this is the current version before use.



CONFIGURATION B

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.010	0.25	.375	9.52	.906	23.01	3.000	76.20
.062	1.57	.438	11.13	.938	23.83	3.313	84.15
.125	3.18	.484	12.30	1.000	25.40	3.500	88.90
.138	3.51	.500	12.70	1.531	38.89	3.750	95.26
.142	3.61	.750	19.06	2.000	50.80	4.500	114.30
.210	5.33	.813	20.66	2.406	61.11		
.266	6.76	.875	22.22	2.960	75.18		

NOTES:
1. Dimensions are in inches.
2. Terminal cover not required.
3. Additional flat washer may be used for terminal seat.
4. Part number MS24172-D1 replaces part no. MS24172-1.
5. This specification sheet takes precedence over documents referenced herein.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.
7. Metric equivalents are given for general information only.
8. Unless otherwise specified, tolerances are ±.062 (1.57 mm).
9. Terminal temperature rise under continuous current conditions, 95°C. Intermediate current to be conducted at 71°C.
10. Shape of relay optional within envelope dimensions.
11. Cadmium or cadmium compounds are prohibited on external hardware.
12. Bolt heads for terminals may extend below their mounting surface.

FIGURE I. Outline drawing (for details see tables I and II) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:49Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and characteristics.

Dash number MS24172-	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1 1/	III	dc	Stud	Bracket - bottom	1.33
D2 2/	III	dc	Stud	Bracket - side	1.33

1/ For Government logistics support MS24172-D1 shall be used in lieu of AN3371-2.
2/ For Government logistics support MS24172-D2 shall be used in lieu of AN3371-1.

TABLE II. Operating characteristics.

PIN MS 24172-	Coil data											Time - (milliseconds maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Bounce			
														Main		Aux	
		Volts 1/	Freq Hz	Res Ω +15% -10	Volts	Amp	Normal 2/	High temp. test	Cont current test					NO	NC	NO	NC
D1	X1,X2	28	dc	10	29	4.5	7.5	11.5	---	3.0	.5	25	18	5.0	---	---	---
D2	X1,X2	28	dc	10	29	4.5	7.5	11.5	---	3.0	.5	25	18	5.0	---	---	---

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	50	200											
Inductive	10	100											
Motor	50	200											
Lamp													
Transfer load													2/
Mechanical life reduced current	100	50											
Mixed loads													

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:49Z
Check the source to verify that this is the current version before use.

Environmental characteristics:

Temperature range	-55°C to +71°C
Maximum altitude rating	50,000 ft
Shock G-level	25 g's
Duration	6-9 ms
Max duration contact opening	2 ms
Vibration - sinusoidal (see table IV)	
Vibration - Random	N/A
High shock	N/A
Acceleration	10 g's

TABLE IV. Vibration levels.

5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1,500 Hz
.08 DA	.06 DA	2 g's	2 g's	

Electrical characteristics:

Insulation resistance, initial:	100 megohms.
After life or environmental tests:	50 megohms.
Dielectric strength (sea level):	2-5 seconds.

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250 V rms	N/A	1,000 V rms	N/A
Aux contacts	1,250 V rms	N/A	1,000 V rms	N/A
All other points	1,250 V rms	N/A	1,000 V rms	N/A

Dielectric strength (altitude): 1 minute.

	28 V dc	115 V ac
Coil to case	500 V rms	N/A
Aux contacts	500 V rms	N/A
All other points	700 V rms	N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:49Z
Check the source to verify that this is the current version before use.

Max contact drop initial:	0.150 volt.
After life test:	0.175 volt.
Overload current	(NO) 1,600 amperes.
Rupture current	(NO) 2,000 amperes.
Duty rating	Intermittent
RFI spec	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Qualification by similarity: See MIL-PRF-6106.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-461

The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Navy - AS
Air Force - 11
DLA - CC

Preparing Activity
DLA - CC

(Project 5945-1271)

Review activity:
Navy - EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at www.dodssp.daps.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:49Z
Check the source to verify that this is the current version before use.